There will be 2 black squares left to tile. They cannot be tiled since a domino has 1 white square and 1 black square.

$\therefore$ The board shown in Figure 5 cannot be tiled. □

We can use other types of pieces besides dominoes in tilings. Instead of dominoes we can study tilings that use identically shaped pieces constructed from congruent squares that are connected along their edges. Such pieces are called **polyominoes**, a term coined by the mathematician Solomon Golomb in the book "Polyominoes."

We will consider two polyominoes with the same number of squares the same if we can rotate and/or flip one of the polyominoes to get the other one. For example, there are two types of triominoes (see Figure 6), which are polyominoes made up of three squares connected by their sides. One type of triomino, the **straight triomino**, has three horizontally connected squares; the other type, **right triominoes**, resembles the letter L in shape, flipped and/or rotated, if necessary.

Example 21: Can you use straight triominoes to tile a standard checkerboard?

Solution: The standard checkerboard contains 64 squares and each triomino covers three squares. Consequently, if triominoes tile a board, the number of squares of the board must be a multiple of 3. Because 64 is not a multiple of 3, triominoes cannot be used to cover an 8x8 checkerboard. □

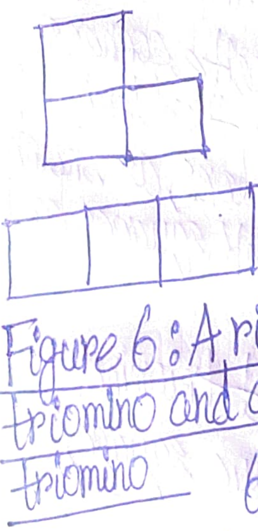

Figure 6: A right triomino and a straight triomino

Example 22: Can we use straight triominoes to tile a standard checkerboard with one of its four corners removed? An 8x8 checkerboard with one corner removed contains $64-1=63$ squares. Any tiling by straight triominoes of one of these four boards uses $63/3=21$